1

$Y_i = \beta_1 X_i + \epsilon_i, i = 1, ..., n$

i

First find method of least squares, find the sum of residuals squared

$$Q = \sum_{i=1}^{n} \epsilon_i^2$$

$$Q = \sum_{i=1}^{n} (Y_i - \beta_1 X_i)^2$$

Take derivative with respect to β_1 and set to 0

$$\frac{dQ}{d\beta_1} = -2 \sum_{i=1}^{n} X_i (Y_i - \beta_1 X_i) = 0$$

Solve for $\hat{\beta}_1$

$$\sum_{i=1}^{n} X_i Y_i = \sum_{i=1}^{n} \hat{\beta}_1 X_i^2$$

$$\sum_{i=1}^{n} X_i Y_i = \hat{\beta}_1 \sum_{i=1}^{n} X_i^2$$

$$\frac{\sum_{i=1}^{n} X_i Y_i}{\sum_{i=1}^{n} X_i^2} = \hat{\beta}_1$$

$$\frac{\sum_{i=1}^{n} Y_i}{\sum_{i=1}^{n} X_i} = \hat{\beta}_1$$

$$\frac{n\bar{Y}}{n\bar{X}} = \hat{\beta}_1$$

$$\hat{\beta}_1 = \frac{\bar{Y}}{\bar{X}}$$

Therefore the OLS estimator for β_1 is $\hat{\beta}_1 = \frac{\bar{Y}}{\bar{X}}$

ii

Prove $E(\hat{\beta}_1) = \beta_1$

$$\hat{\beta}_1 = \frac{\hat{Y}}{\hat{X}} = \frac{\sum_{i=1}^{n} Y_i}{\sum_{i=1}^{n} X_i}$$

$$E(\hat{\beta}_1) = \frac{\sum_{i=1}^{n} E(Y_i)}{\sum_{i=1}^{n} E(X_i)}$$

$$E(\hat{\beta}_1) = \frac{\sum_{i=1}^{n} \beta_1 X_i}{\sum_{i=1}^{n} X_i}$$

$$E(\hat{\beta}_1) = \beta_1 \frac{\sum_{i=1}^{n} X_i}{\sum_{i=1}^{n} X_i} = \beta_1$$

Therefore the estimator is unbiased

2

i

```r
data = read.csv("data/slumptest.csv", sep=",")
model = lm(CompressiveStrength ~ Water, data=data)
summary(model)
```

```

Call:
lm(formula = CompressiveStrength ~ Water, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-19.3590  -5.4508  -0.9858   4.6896  18.8254 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 55.48241    7.39784   7.500 2.54e-11 ***
Water       -0.09861    0.03733  -2.642  0.00956 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.618 on 101 degrees of freedom
Multiple R-squared:  0.06464,	Adjusted R-squared:  0.05537 
F-statistic: 6.979 on 1 and 101 DF,  p-value: 0.009557
```

```r
print(paste("Beta0 =" , model$coefficients[1]))
```

```
[1] "Beta0 = 55.4824064174304"
```

```r
print(paste("Beta1 =" , model$coefficients[2]))
```

```
[1] "Beta1 = -0.0986112998623879"
```

ii

If there is no water content then the compressive strength of concrete is on average 55.482.

On average, the compressive strength of the concrete decreases by 0.09861 per unit increase of water content

iii

```r
vcov(model)
```

```
            (Intercept)        Water
(Intercept)  54.7279850 -0.274712638
Water        -0.2747126  0.001393292
```

iv

```r
confint(model, level=0.95)
```

```
                 2.5 %     97.5 %
(Intercept) 40.8070888 70.1577241
Water       -0.1726577 -0.0245649
```

Thus we are 95% confident that when the water content is 0 then the compressive strength of the concrete is between $[40.807, 70.1577]$

Thus we are 95% confident that the compressive strength of the concrete increases on average $[-0.1726577, -0.0245649]$ per unit increase of water content.

v

$H_0 : \beta_1 = -1, H_1 : \beta_1 < -1$

$\alpha = 0.1$

Using the t-test:

$t^* = \frac{\hat{\beta}_1 - c}{s(\hat{\beta}_1)} = \frac{-0.098611+1}{\sqrt{0.001393292}}$

$t^* = 24.1485568675$

From summary, $df = 101$, and from above $\alpha = 0.1$ so the critical t value is -1.29 (left tailed)

Since this is left tailed, any t value greater than -1.29 is rejection. Since $24.1485568675 > -1.29$ we reject H_0.

Thus using the t test we can conclude that the engineer is wrong and the slope is not less than -1.

vi

```r
predict(model, data.frame(Water=230))
```

```
       1
32.80181
```

vii

```r
predict(model, data.frame(Water=230), interval="predict", level=0.95)
```

```
       fit      lwr      upr
1 32.80181 17.42295 48.18067
```